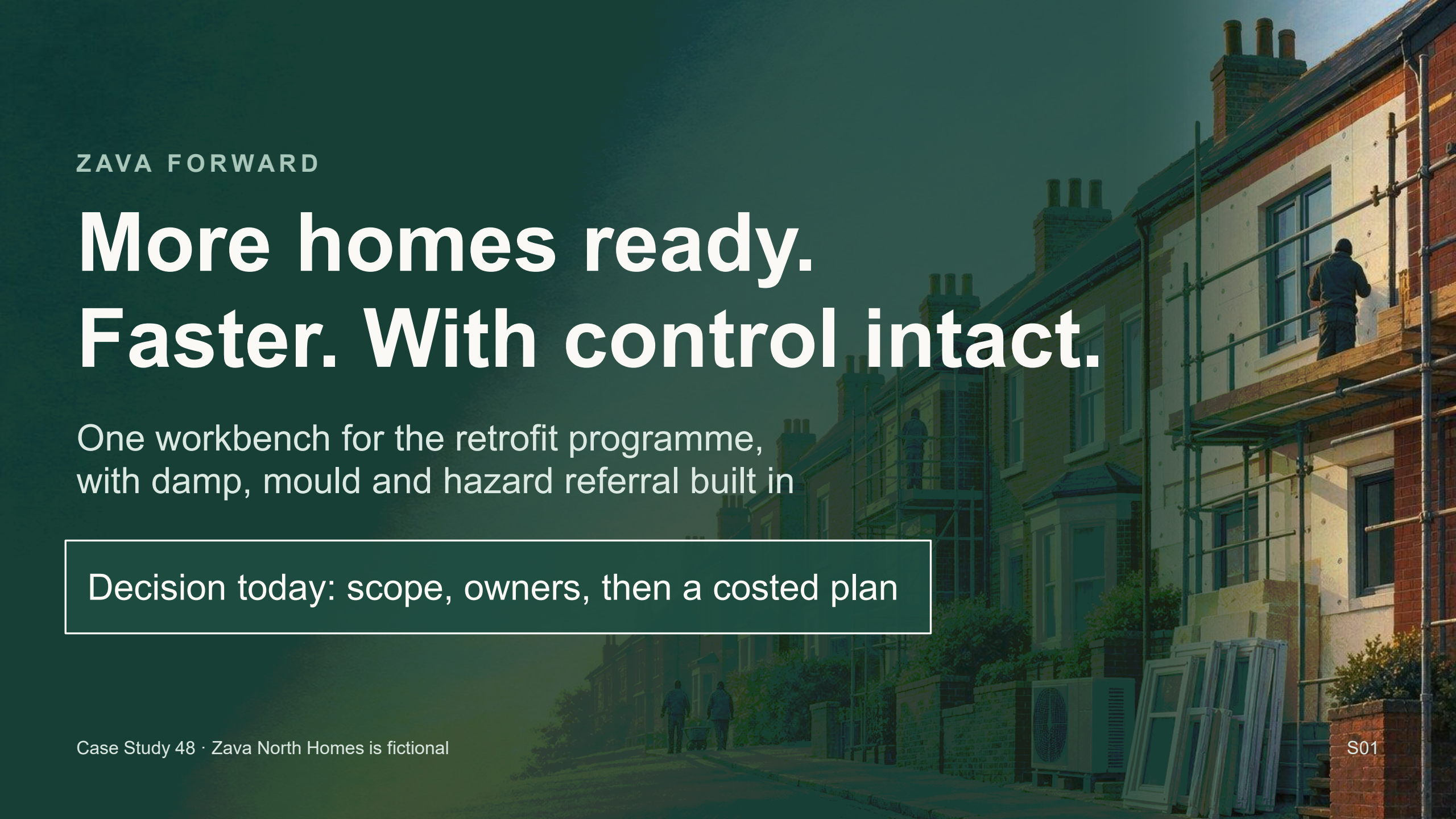


ZAVA FORWARD

More homes ready. Faster. With control intact.

One workbench for the retrofit programme, with damp, mould and hazard referral built in

Decision today: scope, owners, then a costed plan

Case Study 48 · Zava North Homes is fictional

ACT 1 · THE CHALLENGE

15,600 homes to lift by 2030. Thousands more to keep safe.

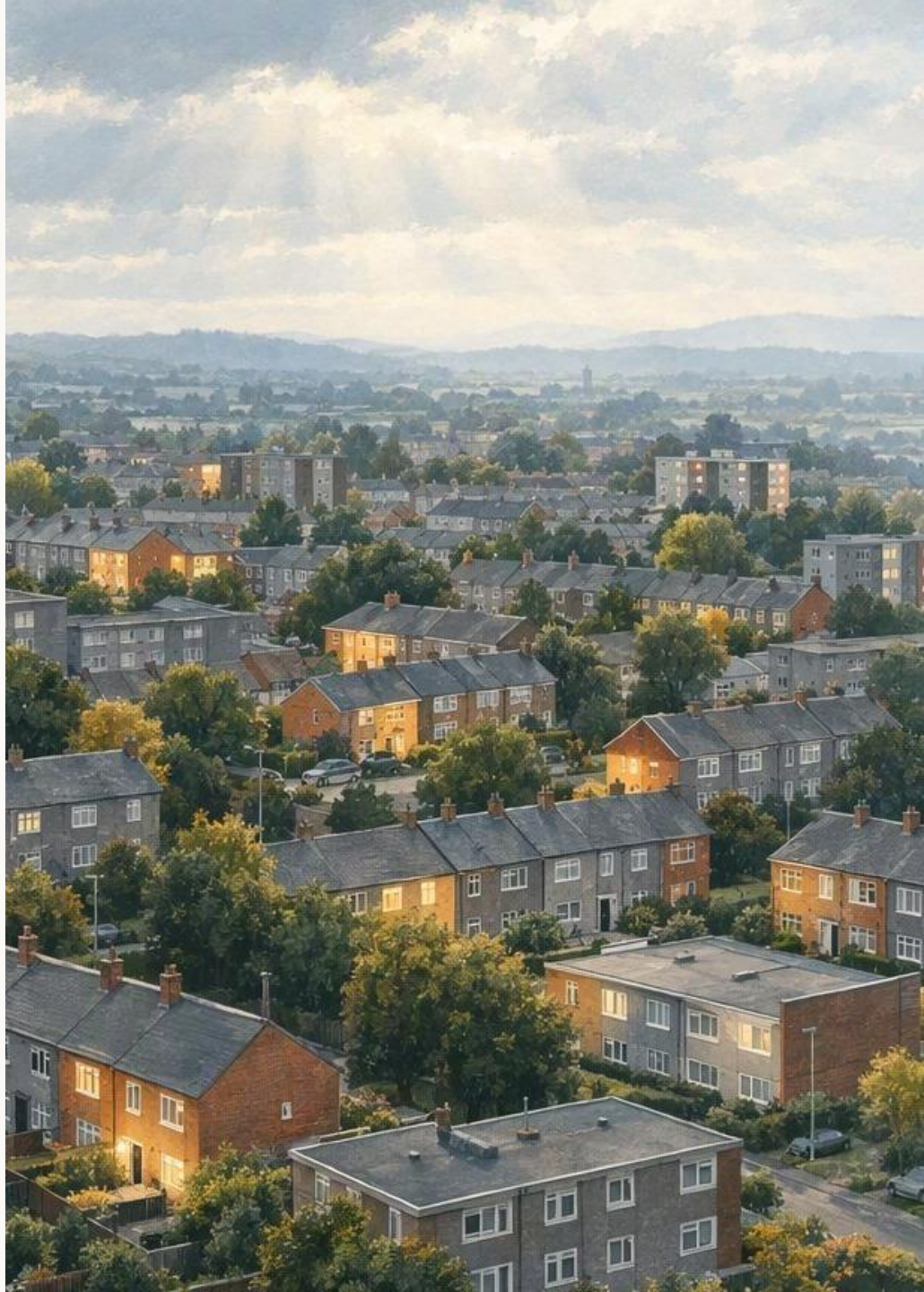
15,600
homes below EPC C
26% of social homes

3,600
homes with damp
6% of housing association homes

22,800
households likely to report condensation, damp or mould
38%

3,210
stage-one complaints a year
53.5 per 1,000 homes

Sources: English Housing Survey 2024-25 · RSH Tenant Satisfaction Measures 2024/25 · national rates applied to 60,000 homes



Three timelines. Different legal and policy status.

DATED SOURCE REGISTER, NOT A FRESH LEGAL CURRENTNESS CHECK

Phase 1 regulations

In-force in local register.
Check current applicability.

Phase 2 guidance

Planned expansion.
Check dates and scope.

Energy / homes policy

Final policy direction.
Commencement not proved.

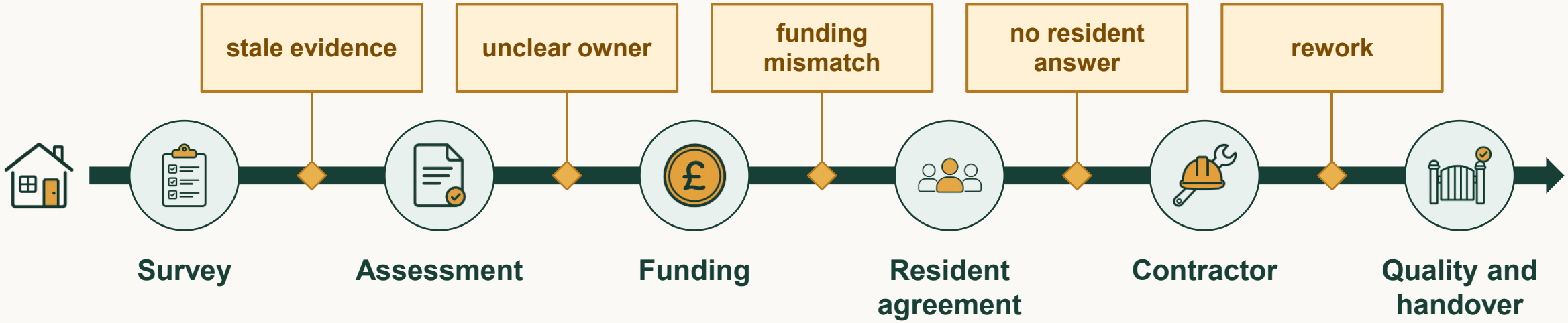
Know the home, the accountable owner and the evidence.

Confirm current sources and named competent review before activating a rule.

Regulatory register: retained 19 Sep 2026 snapshots / register updated 15 Oct | synthesis 29 Oct, no legal approval.

Sources: MHCLG Awaab's Law guidance (Jul 2026) · energy efficiency response (Apr 2026) · Decent Homes policy (Jan 2026)

Homes don't stall on ambition. They stall between teams.



Evidence, funding, capacity and resident choice must align

More starts are not the same as more accepted homes

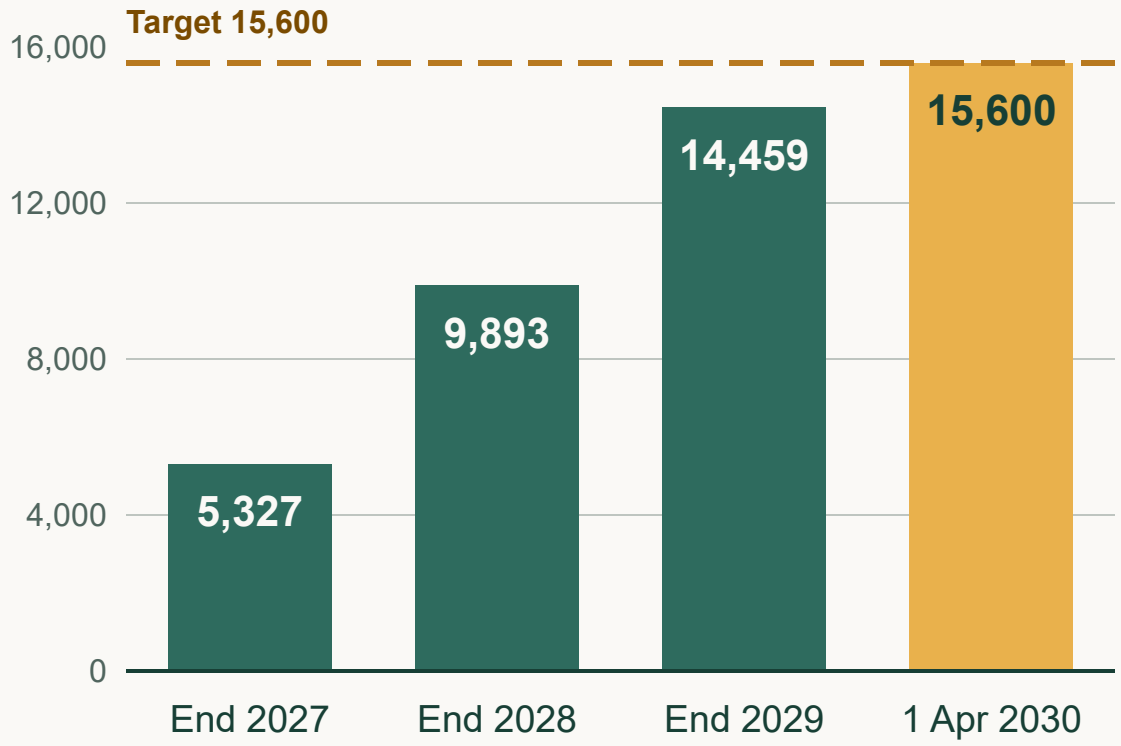
A conflicting record should stop one approval, not all useful work

Sources: exercise journeys, role research and agent-request challenge

Forecast: every Zava home at EPC C by April 2030.

FORECAST

Homes at EPC C, cumulative from Nov 2026



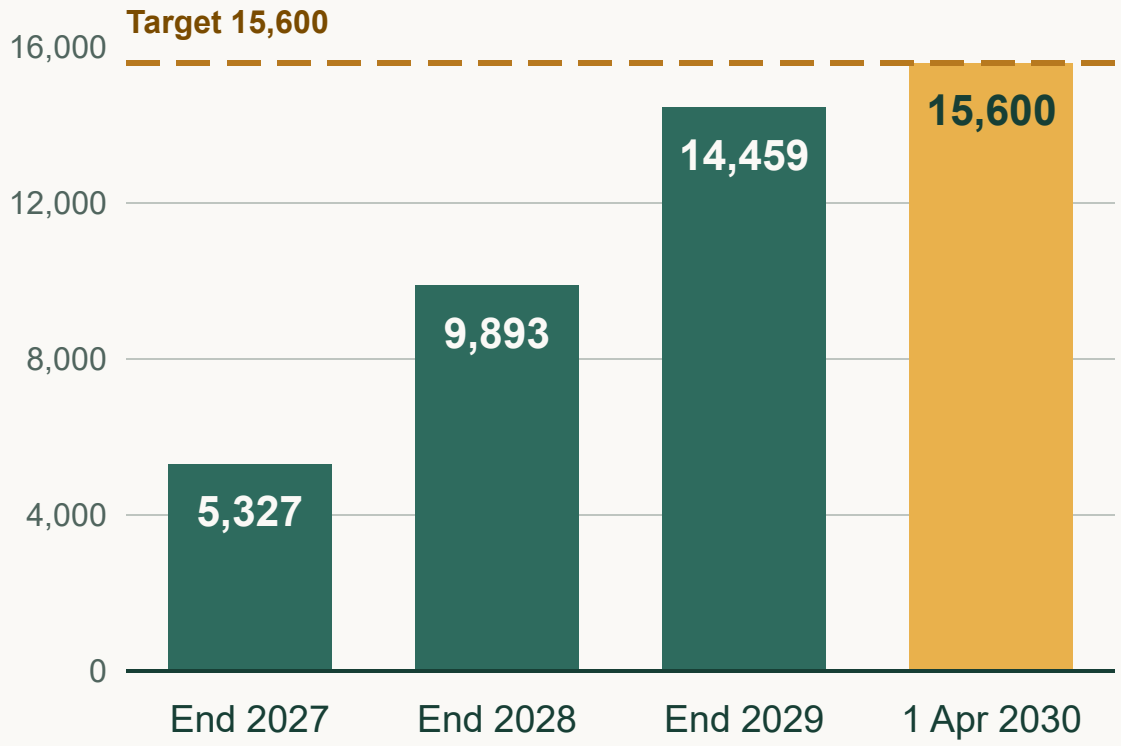
15,600 homes · 41 months

88 homes a week

Forecast: every Zava home at EPC C by April 2030.

FORECAST

Homes at EPC C, cumulative from Nov 2026



15,600 homes · 41 months

88 homes a week

GBP 95m
works benchmark

GBP 47.4m
conditional grant potential

USD 6,873
whole-cohort inference scenario

Forecast: 15,600 homes; E21: assumed workload + secondary price (23 Oct 2026). Grant subject to eligibility, allocation, award and claim.



ACT 2 · DISCOVERY

Six weeks of product work sit under these slides.

What you see: this deck

480 commits	**233** merged pull requests	**6** architecture decisions	**142** requirements
196 acceptance criteria	**34** research documents	**28** journey maps	**189** backlog items

Since 11 Sep 2026
one architect, GitHub Copilot and automation

Sources: public repository jmservera/ama at 140d8f4 (23 Oct 2026) · BRD v0.19.0 · PRD v0.18.0 · AMA Plan board

From the brief to bounded design.

Brief + sector evidence	Role research instruments	Journey hypotheses
Case facts, official context and unknown company details	8 in the autonomy challenge; 9 in the later journey atlas	27 companion maps + 1 atlas map Actions, needs, pains and gains

What the synthesis suggested -> what we designed -> what to test

Pain: repeated story Design: scoped case + owner Test: effort and disclosure	Pain: unowned handoff Design: receiver + chase Test: rejection and delay	Pain: changed evidence Design: version-bound review Test: affected reviews only

Structured desk research, not customer interviews. Pains and gains remain hypotheses to validate.

Sources: original brief; sector evidence; role challenge; journey atlas; DT Coach decision trace (30 Oct).

ACT 2 · DISCOVERY

We challenged the brief to make it shippable.

BRIEF SAID → Five autonomous agents, autonomy L2-L4

WE BUILT

Five bounded assistants on one proposal service

Why: 8 role research instruments concluded none of the five should run end to end.

BRIEF SAID → Fix the spreadsheets

WE BUILT

Data ownership and workflow instrumentation first, then AI

Why: slow delivery is a coordination problem that spreadsheets hide.

BRIEF SAID → AI makes decisions

WE BUILT

AI prepares, people decide

Why: consent, safety and money need a named human.

BRIEF SAID → Resident portal

UNDER ASSESSMENT

Optional portal; assisted routes stay

Test fewer calls and staff interruptions (#441).

Sources: agent-request challenge (E03) · ADR 0002 and ADR 0003 (E06, E08, E09)

S08 · Answers: B03 · B14

Every prototype taught us something we kept.

20 Sep
PR #1 requirements review
open decisions made explicit

6 Oct
review of 11 perspectives, 9 filters, 8 cases
inconsistent owner labels fixed in requirements

11 Oct
PR #58 real AI, Azure SQL and Blob
decision model + generative calls work

23 Oct
PR #433 Tokenomics
AI is managed by cost per accepted outcome



PR #19
four role-based perspectives
roles need their own queues

8 Oct
MVP replaces prototype
one shared workbench

14 Oct
five assistants, real proposals, 73 SQL tests, survived a restart
the proposal service holds

From a requirements review to five real AI assistants in 25 days

Owner feedback reshaped the screen: my cases, next owner, plain status

Every lesson became a requirement, a decision or a test

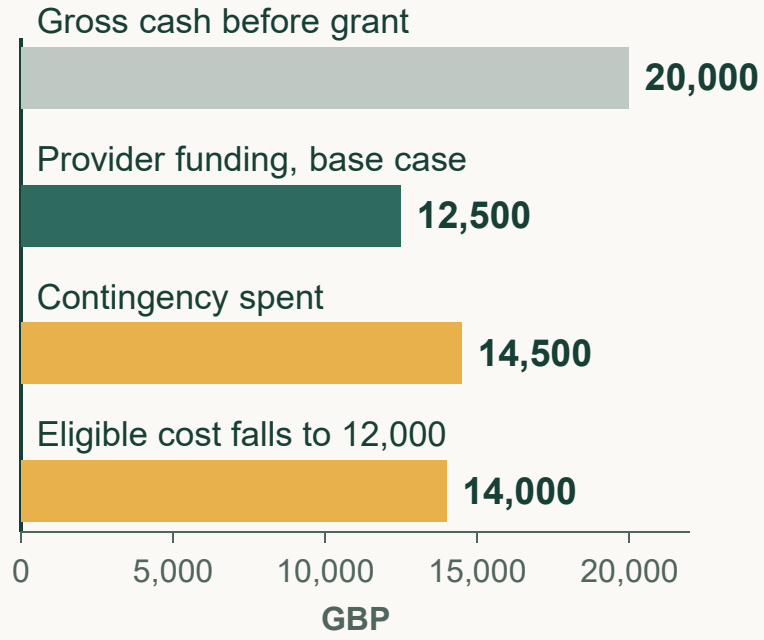
Sources: PRD revision history and learnings · exercise addendum · prototype learnings review · pull requests #1, #19, #58, #433

WHAT-IF

We stress-tested the economics before we wrote the product.

20 scenarios
five agents, a programme supervisor and a damp and mould loop

ONE HOME · exercise fixture (fictional)



THE PROGRAMME · 15,600 homes

WORKS

GBP 6,082/home historical HA benchmark — **94.9m**

By EPC band — **£107m**

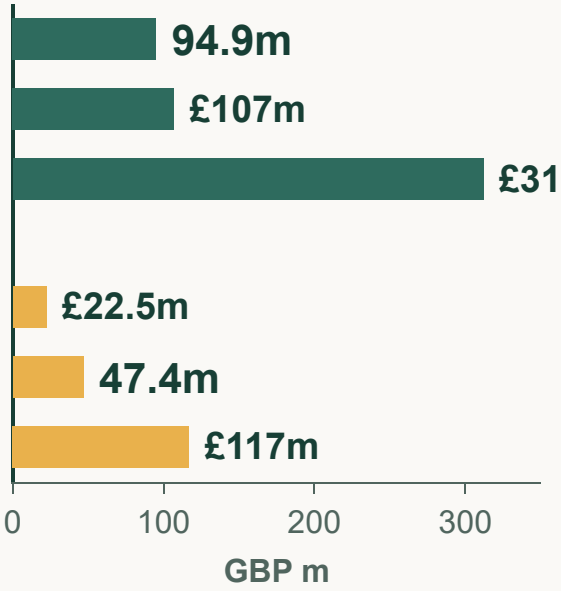
Deep-retrofit fixture (GBP 20,000 a home) — **£312m**

GRANT SCENARIOS

Proportional wave 3 share — **£22.5m**

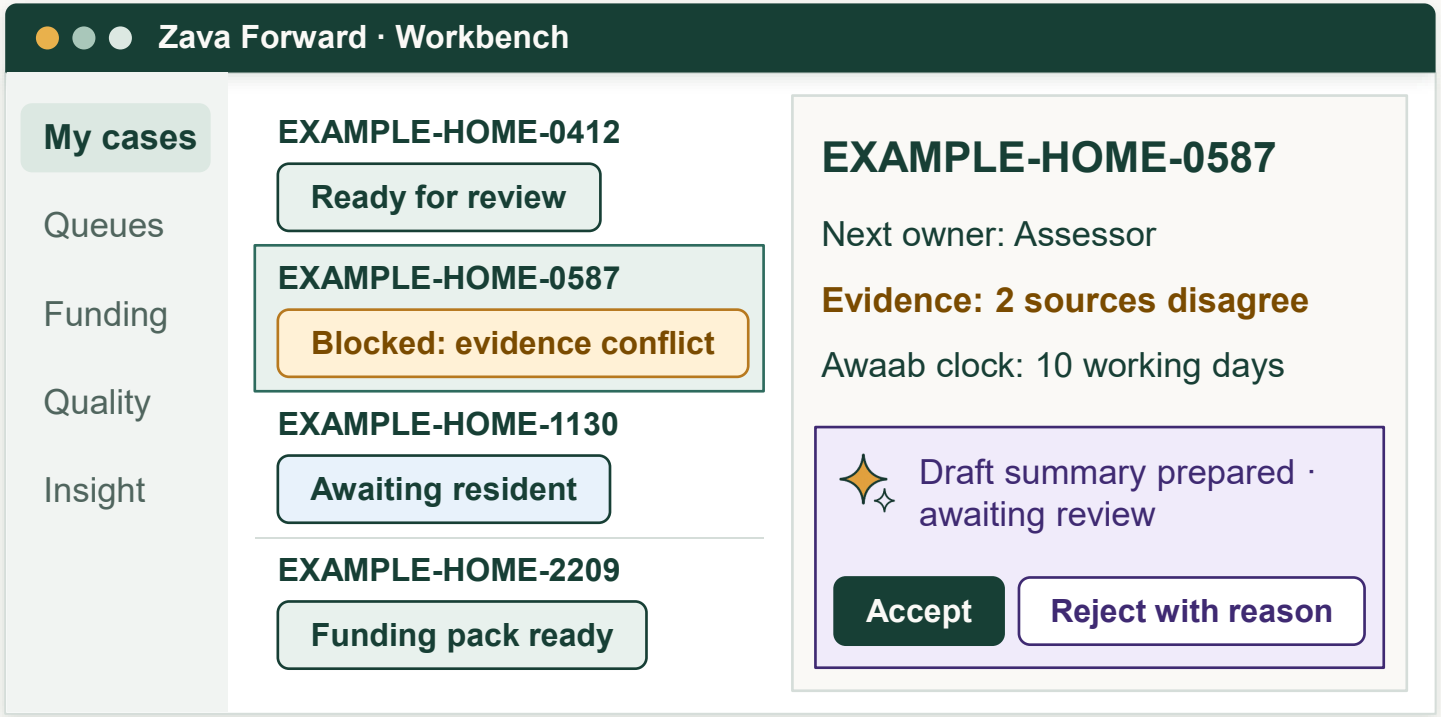
Conditional GBP 3,041/home (50% share) — **47.4m**

Full £7,500 cap — **£117m**



E21: dated benchmark, not a quote. Grant assumes eligible homes/costs, allocation, award and claim. GBP 20,000 fixture remains separate.

Meet the Workbench: every home, one owner, one next step.



My cases · next owner · plain status · bounded help

Delivered in the MVP: work-queue signals, controlled bulk edits, hazard referral, grant states, forecasts

Every decision sits on one evidence trail

Product concept view based on delivered work queue (#77), case controls (#78) and AI suggestions (#106). Fictional data.

Zava Forward: a resident's retrofit journey

Desk-research hypothesis; direct resident validation pending

	01 Contact	02 Survey	03 Decide	04 Installation	05 Live with it
Human role	Housing	Assessor	Resident	Programme	QA / Housing
Do / need	Call / be heard	Plan visit	Choose works	Keep routines	Use / query
Pain (hyp.)	Story repeated	Needs missed	Rushed choice	Disrupted day	Concern lost
Gain (prop.)	Clear contact	Visit support	Pause / defer	Agreed visit	Owned help

Phone / letter / assisted visits; independent QA; urgent safety -> human repairs.

Brief + official sources > 9 role instruments > 28 maps > bounded design

Sources: resident journey atlas and DT Coach source-linked map. Editable Excalidraw + text equivalent retained.

What we built: one accountable workbench.

Case + evidence ➡ **Owned next action** ➡ **Separate decisions**

Work stays owned

Role and case scope
Sent is not accepted
Exceptions stay visible

Changes stay traceable

Source and scope versions
Reopen dependent reviews
QA, handover and cash differ

AI stays bounded

Sourced drafts, human review
Rules own calculations
Manual work remains usable

CURRENT FOUNDATION | Workbench + API + SQL/Blob + rules + bounded AI + read-model reporting

Historical operator exercise: AI proposals persisted in SQL across a restart. Current acceptance and benefits remain unproved.

Sources: current guides; dated operator proof. Current acceptance and value remain separate.

Adopt without a big-bang replacement.

Preview the source

Declared workbook structure
Row/source lineage
Original file unchanged

Review before change

Conflicts and quarantine
Separate human review
No guessed identity joins

Keep familiar outputs

CSV / one-way Excel
Read-only comparisons
Manual and assisted routes

Existing file/CLI tooling; the web import journey is not accepted.

Owners must accept mapping, reconciliation and rollout. A live enterprise connector is a separate decision.

Sources: importer/API guides. Existing CLI/file tooling, not an accepted web import journey.

Five assists. One workflow. People decide.

Stock	Assessment	Funding	Works + residents	Quality
Explain sourced stock evidence	Prepare a sourced survey pack	Draft funding evidence	Suggest slots; draft updates	Assemble evidence for independent review

Rules enforce

Ranking, arithmetic, rights and workflow gates

Decision 1 proposes

Select or abstain.
Must beat a parser.

Generative AI drafts

Sourced narrative;
a person checks the meaning.

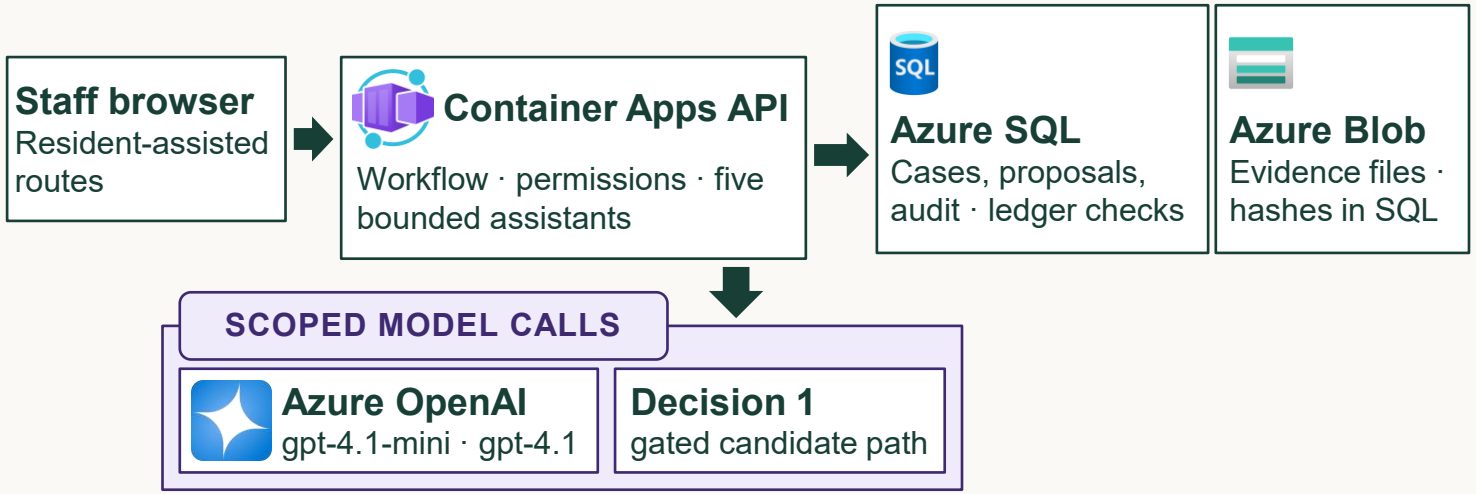
Not selected: model-led approvals or ranking; five autonomous deployed managers.

People retain consent, technical sign-off, Finance approval and quality acceptance.

29 Oct source synthesis | current live acceptance and incremental benefit remain unproved.

Sources: AI placement view (E04, E09) · AMA Plan #104, #105, #106, #107, #214

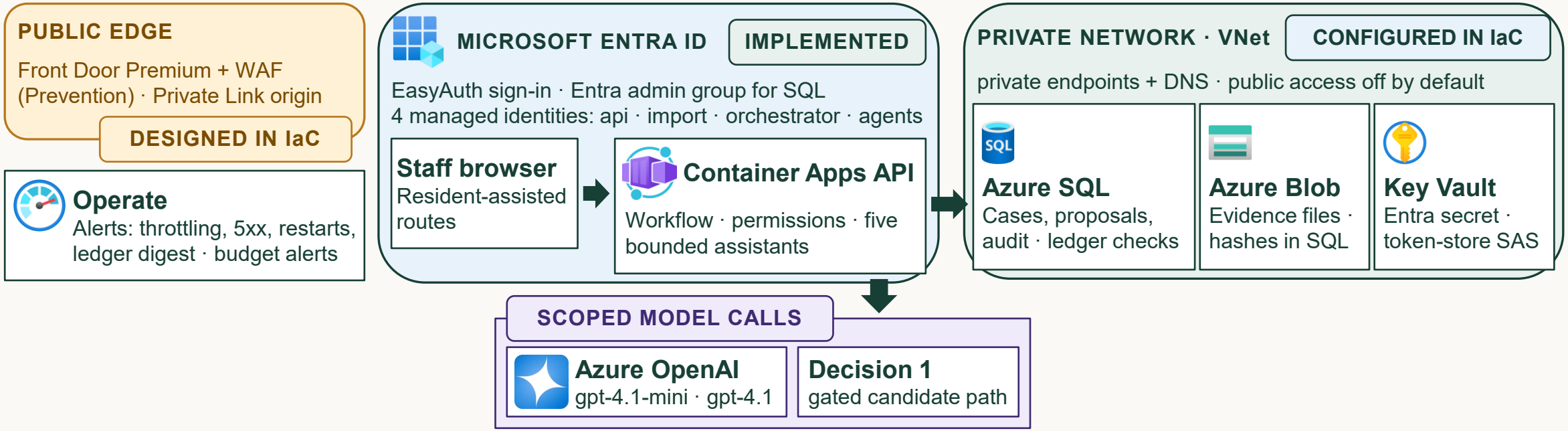
Secure by design: every home record sits inside three boundaries.



Sources: infra/main.bicep · architecture README · ADR 0002 decision 6 · PRD NFR-003 · BRD no-self-approval rules · AMA Plan #80, #102, #143

Filled arrow = request path · tinted region = security boundary · chip = delivery status

Secure by design: every home record sits inside three boundaries.



Current staff sign-in and Graph acceptance remain outstanding.

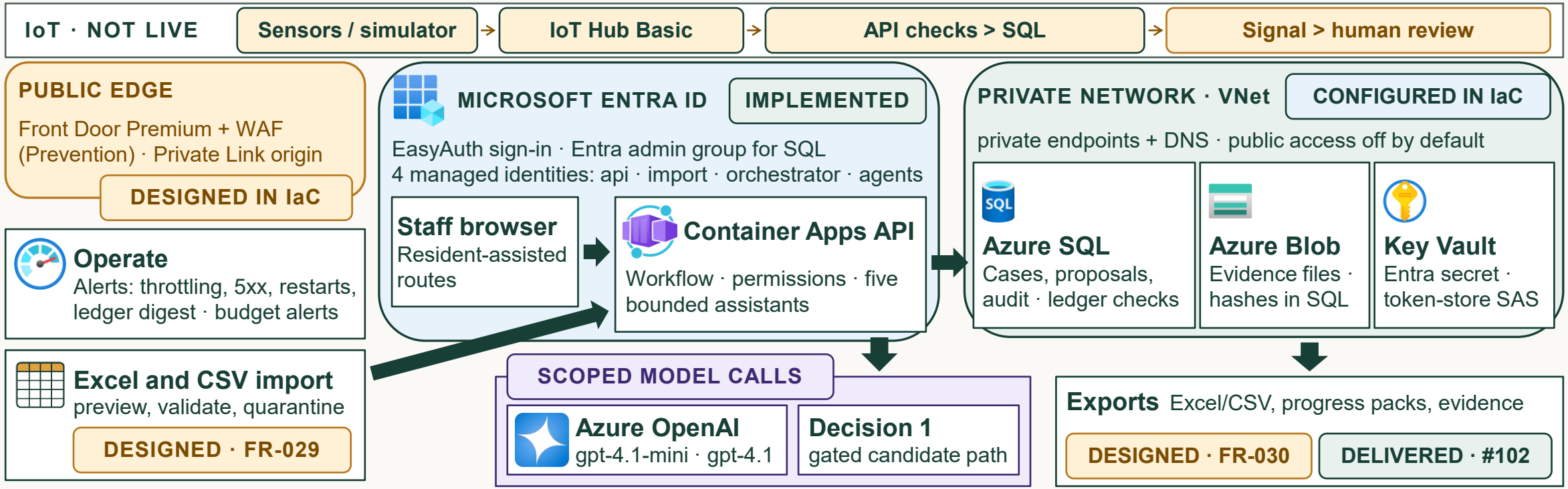
Network: SQL and Key Vault on private endpoints · managed identities · no SQL logins

Sources: infra/main.bicep · architecture README · ADR 0002 decision 6 · PRD NFR-003 · BRD no-self-approval rules · AMA Plan #80, #102, #143

S16 · Answers: B10 · B19 · B22

Filled arrow = request path · tinted region = security boundary · chip = delivery status

Secure by design: every home record sits inside three boundaries.

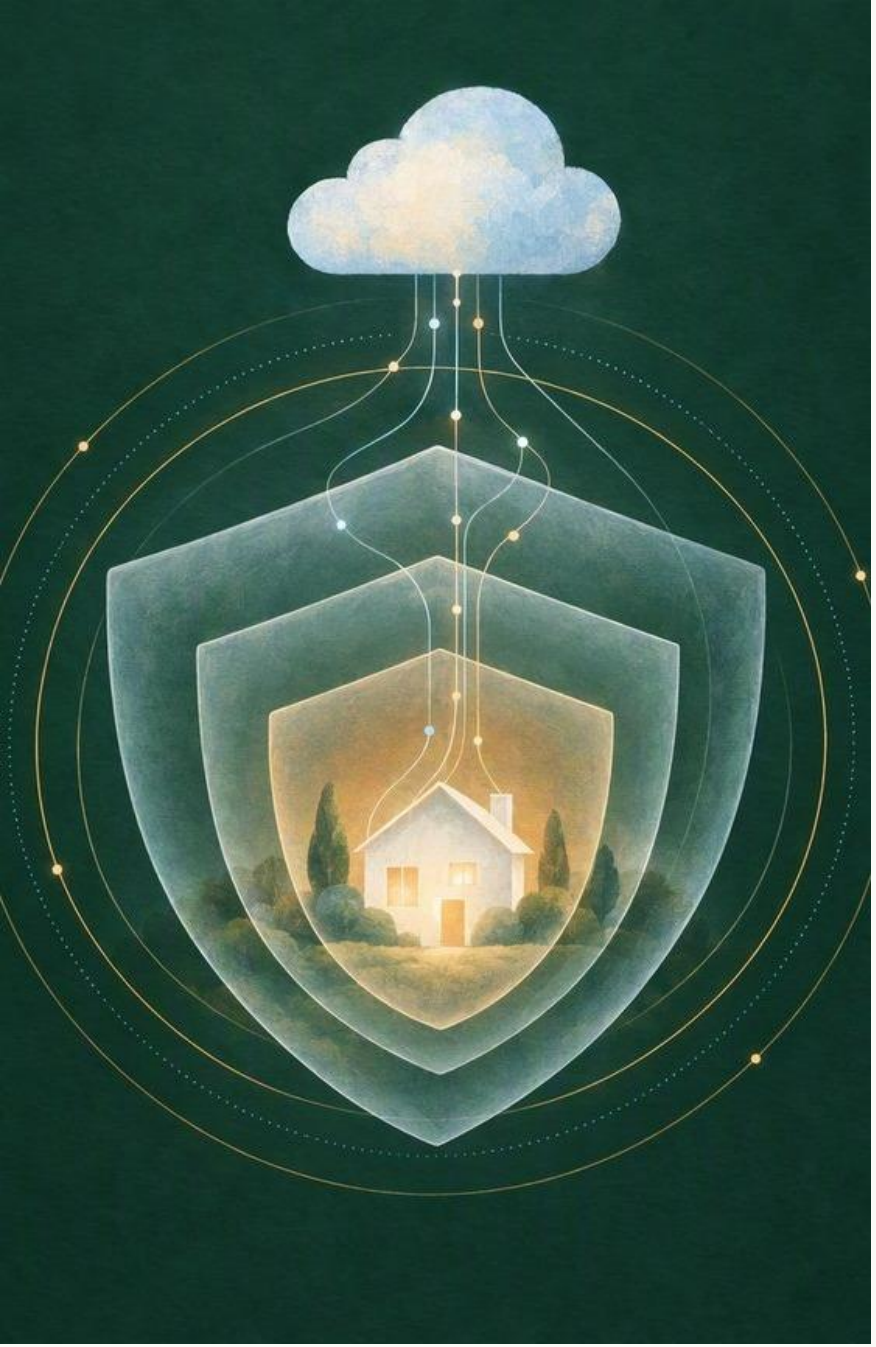


Current staff sign-in and Graph acceptance remain outstanding.

Network: SQL and Key Vault on private endpoints · managed identities · no SQL logins

IoT: monitor after retrofit; no diagnosis. Excel/M365 stay. Portal: assess #441.

Sources: ADR 0002/0003 · infra/main.bicep · PRD NFR-003 · #441 (assessment) · AMA Plan #80, #102, #143



ACT 4 · BUILT TO BE TRUSTED

Seven checkpoints between an AI draft and a real-world decision.

The model prepares. The system checks. A person decides.

1. Signed-in identity (Entra)
2. Role and case scope
3. Permitted evidence only, with source and version IDs
4. Bounded model call, finite allowance and expiry
5. Schema, citation and freshness checks: stale or error stops
6. Human review with current permissions: accept or reject with a reason
7. Separate domain decision and audit trail (consent, quality, Finance)

Failure never becomes approval: the work falls back to its owner
Hazards bypass the AI path entirely
Unknown, stale, disabled or over-allowance → owned manual work · hazards go straight to people

Sources: controlled AI flow (diagram 04, E09) · ADR 0002 · AMA Plan #105, #106

S17 · Answers: B07 · B16 · B21

ACT 4 · BUILT TO BE TRUSTED

Model inference. Measure accepted value.

INFERENCE PLANNING SCENARIO

USD 0.44/home

USD 6,873 / 15,600 homes

20 tasks x 13,768 tokens
x USD 1.60/million

Accepted cost

Unknown

No accepted-outcome
unit cost established

Include review effort
and assurance

Double workload

USD 0.88/home

Inference only. Full project TCO and accepted-outcome cost remain unknown.

IN PLACE

finite calls, tokens and expiry per authority
usage captured on the assistant path
Azure budget alerts

NEXT

proposed money budgets per task and period
per-request call and depth limits
accepted-outcome cost: not yet measured

E21: assumed workload; secondary price, 23 Oct 2026, not current verified price. E19: usage only. E05: controls proposed; #435 not accepted.

S18 · Answers: B05 · B25 · B26 · B27

Everything But the Software

Author retrospective / exploratory single case / non-causal

183 merged PRs

15-24 Oct 2026

36,755 log words

added 15-22 Oct 2026

THE OUTCOME THAT CHALLENGED THE OUTPUT

The author could no longer use the replacement.

Output counts do not establish usable delivery or causation.

Feature -> Tests -> Use journey -> Keep / cut / tidy

Bounded paper test: can another agent supply reflection? Paused; registration pending; no results.

Current source: Everything But the Software, draft + author clarification, 25 Oct 2026; Table 2 defines each window.

Future: connected history and better agent context.

Microsoft Fabric

Power BI

FUTURE OPTIONS ONLY | Neither is part of the current Zava Forward product.

Portfolio history, reused

Joined evidence across time
Shared reporting definitions and measures

Fabric ontology (preview)

Home -> Assessment -> Measure -> Evidence
Governed definitions, links and metrics

60,000 homes

Fictional stock, not resident accounts

980 cases

Separate retained programme lineage

43.8M readings

Synthetic daily history over 730 days

Why later: prove value, curated data, permissions, preview fit and full cost first. Scale alone proves no platform need.

Sources: Fabric assessment + 30 Oct ontology follow-up. Preview; no integration or value gain proven.

Prove the need. Choose the smallest next step.

PRESERVE THE CURRENT FOUNDATION | evaluate a need, not a service count

1 Prove the current journey

Access, recovery and review
Resident and staff effort
Relevant actual evidence

2 Compare portfolio insight

SQL/read-model vs Fabric/PBI
Same cohort and history
Quality and complete cost

3 Test richer agent context

Current context vs ontology
Checked answers and refusals
Stale/forbidden link tests

Agree owners. Cost the plan; authorize separately; expand only on evidence.

Sources: measurement design and acceptance boundaries. Separate execution authority required.

See it work: one home, one exception, one accountable next step.

1



Watch the evidence conflict find its owner

2



Watch what AI drafts, and what only a person can approve

3



Watch the record keep the next action



Ten-minute demo • route confirmed at preflight: live exercise or labelled evidence walkthrough

Sources: demo runbook (E07, E13)

ZAVA FORWARD

More homes ready, with control intact.

1. Agree the validation gate
2. Name the owners
3. Review a costed plan



APPENDIX · NAVIGATION ONLY · END OF THE SPOKEN ROUTE

Questions we expect, and where the answers are.

STRATEGY AND VALUE

B01 · 28 Why not fix the process and hire more people?
B08 · 35 Will this double delivery and capture GBP 40m?
B09 · 36 What is the per-home case and cash exposure?
B17 · 44 When can we go live, and who owns next?
B20 · 47 Why Fabric later? Value, cost and alternatives

METHOD AND READINESS

B12 · 39 Validated with real residents and staff?
B13 · 40 How did requirements become the backlog?
B15 · 42 What must happen before a pilot decision?

COST AND TOKENOMICS

B04 · 31 When does Decision 1 beat simpler rules?
B05 · 32 What does the technology cost today?
B25 · 52 Tokenomics unit: can the AI bill run away?
B26 · 53 Why not cache everything, use the cheapest model?
B27 · 54 What stops chat and agent sprawl?

ARCHITECTURE AND AI

B02 · 29 Replacing or integrating our housing system?
B03 · 30 Why not exactly the five agents requested?
B10 · 37 Why not deploy each agent independently?
B11 · 38 What proves the AI is real and useful?
B14 · 41 Is it agentic if so much is deterministic?

TRUST, RISK AND PEOPLE

B06 · 33 How do residents keep choice, incl. offline?
B07 · 34 What stops humans rubber-stamping AI?
B21 · 48 How do you prevent hallucination and injection?
B22 · 49 Can data leave the UK or train a model?
B23 · 50 How will you know it is fair and accessible?
B24 · 51 How do policy, law and funding stay current?

OPERATIONS AND DEMO

B16 · 43 What if the model, cloud or deployment fails?
B18 · 45 Can you demo a new live action now?
B19 · 46 Sensors: can AI diagnose damp or mould?

Why not fix the process and hire more people instead?

OPTION 1

Manual / process only

Process improvement is the baseline

TRADE-OFF

Additional staff or contractor capacity may still be necessary

STAFFING PLAN SET IN PILOT

OPTION 2

Workflow + rules

Workflow must beat the baseline

TRADE-OFF

If rules and templates deliver the better accepted-task result, retain them

COMPARISON 1 · READY TO RUN

OPTION 3

Workflow + bounded AI

AI must beat the same workflow without AI

TRADE-OFF

If rules and templates win, report the negative AI finding

COMPARISON 2 · READY TO RUN

ALL OPTIONS **Physical delivery capacity still limits throughput**

Sources: E02 · E10

Is this replacing our housing system or integrating with it?

CURRENT DEMONSTRATOR

Controlled workbench

Demonstrator of coordination and evidence contracts

Not a replacement decision

Housing and repairs integration is the next step

INTEGRATION ROADMAP

- **Housing system**
- **Repairs system**
- **Configured or designed M365 path**

Integration scope, authority and migration need separate discovery

PRODUCTION APPRAISAL MUST IDENTIFY

Authoritative systems · field ownership · APIs · event and reconciliation contracts
Security · retention · the manual service route

Why did you not build exactly the five agents we requested?

Five requested responsibilities are preserved

Autonomy is bounded by decision rights and risk

Five assistants ≠ five independent runtimes

	AI PREPARES	PEOPLE DECIDE
Responsibility	**Bounded help (unreviewed proposal)**	**Human decision owner**
Stock prioritisation	Prioritisation rationale	Approved policy and human review
Assessment	Assessment preparation	Qualified reviewer
Funding	Funding evidence	Finance submits
Coordination	Coordination draft	Resident agrees
Quality	Quality evidence	Independent professional accepts

LATER STAGE

One API hosts the current assistants; independent queue workers remain deferred

Sources: E01 · E03 · E06 · E09

Decision 1 must beat the simpler baseline.

Rules / template

Exact labelled fields
Arithmetic and ranking
Deterministic gates

Decision 1

Supplied candidates
Binary support screening
Select or abstain

Generative model

Sourced narrative
Permitted evidence only
Human checks meaning

Same cases. Held-out quality. Total human effort. Latency. Full cost.

A successful call proves integration, not lower cost, faster work or calibrated accuracy.

29 Oct synthesis | Decision 1 behaviour specification, model-fit research and current evaluation limits.

Sources: E04 · E05 · E15

What does the technology cost today?

USD 700 owner-set October all-in forecast ceiling

Execution gates remain; authority expires 25 October 16:02 UTC

Refresh post-change forecast; outside-group cost remains unknown

Amount	Scope	Class	Status
USD 700	October all-in forecast ceiling, owner decision 22 Oct	Owner-set ceiling	Paid tranches blocked; expires 25 Oct 16:02 UTC
USD 242.36	Accrued 11-22 Oct, resource group ama	Observed, partial	Estimated until invoiced
USD 535.33	Pre-change scenario	Historical forecast	Superseded for authorisation; refresh
USD 450/month	EXA-012, 10 Oct	Historical allocation	Not current authority
Outside-group spend	Outside resource group	Unknown	Never zero by assumption

DATED LEDGER
NOT PRODUCTION TCO

Future project appraisal
P + USD 6,872.9856
Per planned home:
(P + USD 6,872.9856) / 15,600
P = non-inference costs for the same period.
P is not yet priced.

Count each cost once. Do not add the October USD 700 envelope and its included inference.

Sources: E11 / E05 / E21. Historical price assumption: 23 Oct 2026; no verified current quote.

How do residents retain choice, especially without digital access?



Optional portal; assisted routes stay

Offer ≠ agreement; silence ≠ consent

Representative authority is task-limited

DESIGN BOUNDARY · HUMAN VALIDATION NEXT

Direct resident research, approved communications and privacy decisions remain necessary

Sources: E02 · E08 · E12

What stops a human from rubber-stamping AI?

Source verification before proposal acceptance

Current permission, evidence and independent authority

Meaningful oversight remains a human acceptance gate

1. Capture

Role + case scope
Permitted evidence
Source/version IDs

AI PREPARES

2. Call

Bounded task
Model + usage
No database lock

3. Validate

Schema + citations
Stale/error = stop
Store as unreviewed

PEOPLE DECIDE

4. Review

Human checks source
Current permissions

Separate domain action

Failure does not become approval

Unknown, stale, disabled, exhausted allowance or unavailable model → owned manual work; hazards bypass AI.

Citation presence is not semantic truth · Proposal review is not consent, QA or Finance

SOURCE VIEW · 23 OCT 2026

Sources: E02 · E09 · E12

Will this double retrofit delivery and capture GBP 40 million?

1 · AMBITION

Original brief ambitions: the outcomes the pilot measures

Double retrofit delivery

Capture GBP 40 million

ORIGINAL BRIEF AMBITION

2 · DEPENDENCIES

Capacity, scope, eligibility and cash all constrain value

Survey, technical, contractor, material, resident and QA capacity

Grant availability and secured cash: confirmed at the funding gate

3 · TEST

Frozen cohorts and attributable comparisons required

Measure accepted handovers and the whole cost of the work

READY TO RUN

4 · OBSERVED RESULT

NEXT MILESTONE

Benefit range set once the baseline, causal limits and sensitivity are agreed

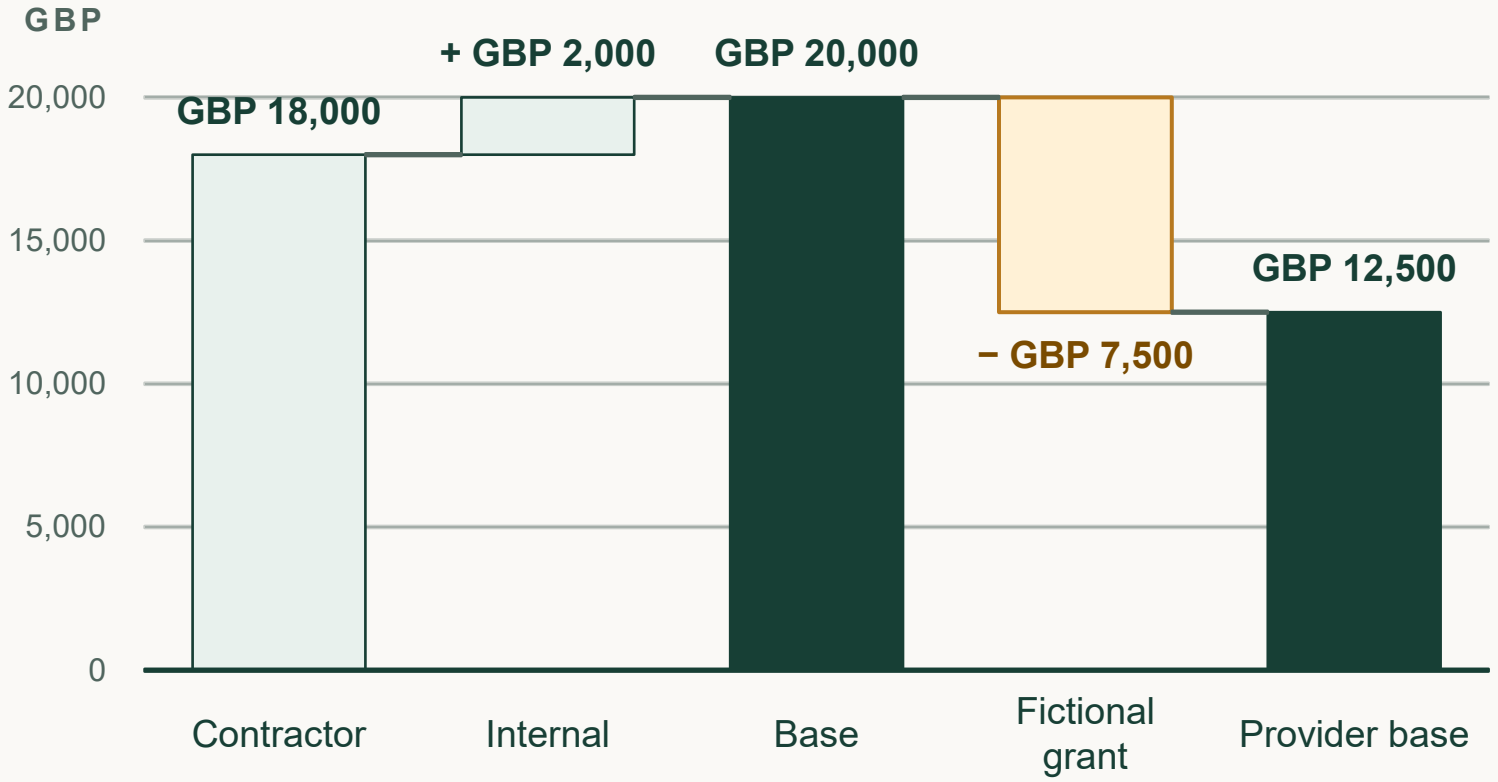
Saved minutes become cash through a Finance-approved mechanism

Sources: E01 · E02 · E10

What is the per-home business case and cash exposure?

Benchmark: GBP 6,082/home
x 15,600 = GBP 94,879,200 works

Conditional: min(50% x GBP 6,082, GBP 7,500)
= GBP 3,041/home; x 15,600 = GBP 47,439,600



FICTIONAL EXERCISE ARITHMETIC

Fictional base GBP 20,000
GBP 22,000 with contingency

Gross cash GBP 20,000 before grant
Fictional receivable GBP 7,500
An award is not cash.

Contingency spent: provider GBP 14,500
Evidence supports claims; not eligibility, award or payment.

E02 / E21: historical benchmark, not quote. Grant assumes eligible homes/costs, allocation, award and claim; Finance/funder review required.

One controlled API now; split only for measured need.

Entra sign-in
API rechecks scope

➡

Container Apps API
Workflow + assists

➡

Azure SQL + Blob
State, audit + evidence

Container Apps, not AKS

Avoid cluster operations for this bounded app.
Fewer service, retry and state boundaries.

SQL state; Blob document bytes

Transactions and versioned reviews in SQL.
Files in Blob; references and hashes in SQL.

Deferred: five deployed workers, queues or a cluster. Split only for measured isolation, scale or recovery need.

ADR 0002 source rationale | no measured platform superiority or current whole-system acceptance claimed.

What proves the AI is real and useful rather than a template?

DATED HISTORICAL EVIDENCE

Invocation
Provider/model/usage and source=ai evidence
Historical real calls and persistence are documented.

≠

NEXT MILESTONE

Semantic acceptance
Semantic quality, critical omissions, unsafe advice and fairness require separate evaluation.

NEXT MILESTONE

Incremental value
Review burden and incremental value require separate evaluation.

LABELLED FALLBACK

Template/replay is clearly labelled as a fallback
The real-inference gate runs on live model calls.

WHAT THE PILOT ADDS

Real invocation proven; calibrated quality and value come from the pilot
Live readiness is re-checked before every run, never assumed.

Sources: E07 · E09 · E04 · E13

Did you validate this with real residents and staff?

No direct resident or staff research has been completed for this design.

Representative five-stage maps / pain hypotheses / desired gains, not measured outcomes

Resident	**1 Hear / report**	**2 Survey**	**3 Choose works**	**4 Appointment**	**5 Live with measures**
	Hypothesis: repeat story / pressure. Gain sought: meaningful choice + owned aftercare.				
Assessor	**1 Prepare**	**2 Arrange access**	**3 Observe**	**4 Write up**	**5 Clarify / revisit**
	Hypothesis: missing records / re-keying. Gain sought: a sourced, traceable assessment.				
Finance preparer	**1 Establish need**	**2 Check route**	**3 Assemble case**	**4 Maintain claim**	**5 Reconcile**
	Hypothesis: conflicting versions / unsupported scope. Gain sought: a defensible pack.				

Next: direct task/comprehension tests, assisted routes, real handoffs and net effort. No endorsement.

Current source: journey atlas, 14 Oct 2026 research / 23 Oct reconciliation; representative hypotheses, not user validation.

How did the requirements become an agent-delivered backlog?

CONCRETE BACKLOG TRACE / retained 23 Oct 2026 deck snapshot

BRD / PRD	Epic #73	Feature #76	Story #77	PR #376

PR evidence is not whole-journey acceptance. HVE has human stops; the coordinator was owner-designed.

Paper's bounded test: can another agent supply reflection between features?

A / Feature only

No extra review budget

B / Consolidate

Review / refactor + tests / note

C / Read-only review

Same extra budget as B

Planned S1: 4 runs per arm / 10 features + CR1. Exploratory case, not causal proof.

PAUSED / registration pending / no results. B/C have extra time vs A; tests and notes also differ.

Current source: Everything But the Software, draft + author clarification, 25 Oct 2026. Historical trace retained above.

Sources: E14 · E16 · E17 · E20

Is this actually agentic if so much is deterministic?

Policy and state

RULES · NO AI

Deterministic orchestration protects controlled state
The application determines eligibility and protects state.

Reasoning and preparation

AI PREPARES

Reasoning and preparation are bounded capabilities
Models reason over permitted context and prepare drafts or flags.

Accountable decision

PEOPLE DECIDE

People retain decisions requiring consent, professional judgement or delegation.

AUTONOMY

Autonomy is task-specific, not a marketing label
Future autonomous actions require an explicit, evaluated task contract and authorised boundary.

Sources: E03 · E06 · E09

What exactly must happen before a pilot or production decision?

NEXT GATE

Current deployment and AI evidence

Current deployment, roles, SQL and real-AI evidence

OWNER ROLE

Technical owner

REQUIRED ARTIFACT

Staff access and role denial, task-bound real inference, SQL persistence on the selected deployment

NEXT GATE

Human and control acceptance

Human, privacy, professional and accessibility acceptance

OWNER ROLE

Professional, privacy, accessibility and safety reviewers

REQUIRED ARTIFACT

Task/comprehension, accessibility, safety/privacy and professional acceptance

NEXT GATE

Value, recovery and release

Measured value, cost, recovery and explicit release decision

OWNER ROLE

Programme and Finance

REQUIRED ARTIFACT

Measured value and cost against the comparator, recovery test, explicit release decision

Sources: E02 · E12 · E13 · E17

What happens if the model, cloud or deployment fails?

Resilience is configured in the architecture; whole-system recovery drills are the next milestone.



Access + secrets

Entra and managed identities
Key Vault secret references
Private backing services



Runtime + releases

Container Apps / pinned images
Registry and CI checks
Controlled rollout + rollback



Observe + budget

Monitor / Log Analytics
Scoped allowance + usage
Budget alerts warn; hard stops are designed.

Evidence required before expansion

NEXT GATE

Genuine staff sign-in + role denial	Restore and recovery drill
Azure SQL persistence	Human task/comprehension checks
Bounded real-model result	Current cost + processing terms

Restart proven; full restore drill is the next milestone

OWNED MANUAL ROUTE

No approval from failure or stale evidence
Owned manual work and independent safety handling
Keep manual and safety routes independent; do not relax private controls to rescue a demo.

Columns are concerns, not request hops · Amber = next milestone

SOURCE VIEW · 23 OCT 2026

Sources: E06 · E07 · E13

When can we go live and who owns the next step?

Dependency-led plan: dates follow the gates

SEQUENCED PLAN

READINESS SEQUENCE	OWNER	
1 Agreed validation scope	Programme	Programme owns scope; Finance owns spend review
2 Refreshed cost	Finance	
3 Deployment prerequisites	Technical owner	Technical and control owners supply acceptance evidence
4 Finite test authorised	Programme + Finance	
5 Dates set	→	LATER, SEPARATE DECISIONS: Commercial pilot; Operational release

Can you demonstrate a new live action right now?

DEMO BOUNDARY

Only pre-authorised, reversible constructed scope
No unplanned role, network, spend or data changes
We will not grant a role, bypass private networking, enable assistant authority, submit a grant or send a resident message on the fly.

DEMO ROUTE, CONFIRMED AT PREFLIGHT

→ **L · Live, preflight passed**
Only within the preflighted, authorised demonstration boundary

→ **E · Recorded walkthrough**
Fallback is visibly labelled evidence, never simulated live proof
Switch once; say exactly what it does and does not prove.

→ **U · Unavailable**
Explain the control instead

× **No silent mock fallback**

Sources: E07 · E13

Device intake is not a diagnosis or safety decision.

SOURCE-ONLY TARGET FLOW | LIVE END-TO-END HUB / SQL ACCEPTANCE UNPROVED

Synthetic device	IoT Hub Basic	Private Event Hub	API + SQL	Human review
Exact binding Stable reading ID	Device boundary One-way intake	Hub-only sender Keyless receiver	Validate + replay Owned DQ tasks	Review signal only No diagnosis

Why Basic: device identity + one-way intake.
No Standard device-control features needed for this exercise.

Hazard reports go straight to human safety.
Neither telemetry nor AI may delay the response.

27 Oct source-only transport / 29 Oct synthesis | synthetic permissions are not real consent or calibration.

Sources: E12

Keep SQL for control. Evaluate Fabric for analytics.

RETAIN: API + Azure SQL + Blob | transactions, ledger and accountable workflow

Alternatives and limits

Keep Excel / M365 as bounded inputs.
No verified low-code or bought-product fit yet;
compare integration and control costs.

No Fabric operational replacement

Fabric SQL lacks ledger creation.
A warehouse is not the write workflow.

Future option: selective Fabric

Reusable cross-source history;
shared measures and portfolio reports.



Compare with SQL views + Power BI.
Require better effort, freshness or total cost.

No purchase or export approved. Access, history and total cost need separate review.

Fabric fit assessment, 29 Oct 2026 | proposed platform advice, not a new ADR, price quote or measured benefit.

How do you prevent hallucinations and prompt injection?

Permitted task context and source/version binding

Schema/citation checks plus independent semantic review

Built to contain vulnerabilities and hallucinations, not to deny them

1. Capture

Role + case scope
Permitted evidence
Source/version IDs

2. Call

Bounded task
Model + usage
No database lock

3. Validate

Schema + citations
Stale/error = stop
Store as unreviewed

4. Review

Human checks source
Current permissions
Separate domain action

UNTRUSTED EVIDENCE

Documents are evidence, not instructions

REVIEW BOUNDARY

Citation presence is not semantic truth

Failure does not become approval

Unknown, stale, disabled, exhausted allowance or unavailable model → owned manual work; hazards bypass AI.

Filled arrow = guarded request/response step, not agent-to-agent talk

SOURCE VIEW · 23 OCT 2026

Sources: E09 · E12 · E07

Can customer data leave the UK or train a provider's model?

Processing geography is set by the deployment type

UK-only inference is a deployment choice (a regional deployment rather than GlobalStandard), confirmed at the security gate.

Verify exact model, deployment type and contract

before real data is processed

- VERIFY Provider
- VERIFY Model and version
- VERIFY Deployment SKU
- VERIFY Routing and geography
- VERIFY Retention
- VERIFY Abuse monitoring
- VERIFY Subprocessors
- VERIFY Applicable agreement

No real resident data or private dataset in this handoff

The handoff contains only public synthetic illustrations and repository-authored summaries, not private datasets or credentials.

How will you know this is fair and accessible?

No-digital route and understandable explanations

Cohort errors, exclusions and unknowns are visible

Engineering checks ≠ resident or WCAG acceptance

	Task success	Comprehension	Errors and exclusions by cohort
Digital route	MEASURED IN PILOT	MEASURED IN PILOT	MEASURED IN PILOT
No-digital route	MEASURED IN PILOT	MEASURED IN PILOT	MEASURED IN PILOT

Inspect errors and exclusions by agreed cohorts without exposing small groups

GUARDRAIL

An AI result that improves average speed while worsening a critical group outcome should not pass the gate.

Conformance and fairness are assessed with real users and a full audit in the pilot.

Sources: E02 · E12 · E15

How will you keep policy, law and funding rules current?

Detected change → reviewed rule change → tests

REPOSITORY CONVENTION · READY TO ACTIVATE

Source

Dated sources with authority classification

Reviewed change item

Affected requirements, rules, journeys, tests

NAMED OWNER

Approved control version

Human approval of a reviewed control update

NAMED OWNER

Regression evidence

Tests and evidence before consequential use

Models may retrieve, compare and flag

They must not decide that a legal duty is satisfied

No silent model update to live controls

Or silently activate a new rule

Dated official-source snapshots; currentness and named legal review required before consequential use.

Sources: E02 · E12 · E18

What is your Tokenomics unit and can the AI bill run away?

Planning: USD 0.440576/home inference; accepted-outcome TCA unknown

Historical secondary price: 23 Oct 2026. Assumed workload, not current verified deployment price.

Observe, then assume

6,884 / 5 = 1,376.8 tokens/run
x 10 = 13,768 tokens/task
x 20 = 275,360 tokens/home

Price inference only

275,360 / 1m x USD 1.60
= USD 0.440576/home
x 15,600 = USD 6,872.9856

Measure accepted TCA

Hosting + review + assurance
Accepted above quality floor
TCA / accepted work: unknown
Double workload: USD 0.881152/home

Inference scenario is not total project cost or accepted value

DESIGNED · NEXT MILESTONE

Existing bounded path

Finite calls / tokens / expiry; partial metering
Not hard currency or request-depth limits.

Proposed: not run

Complete attribution; task/period money stops
Request/depth bounds; billing reconciliation.

Tokenomics framing: our working method, refined with Finance

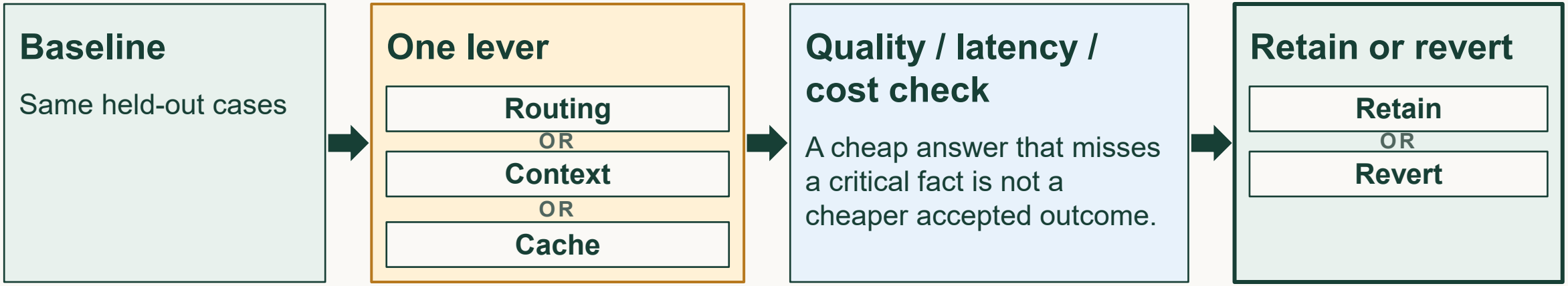
RECOVERY: 24 OCT 2026

Sources E19 / E05 / E11 / E21. #435 not accepted; no saving, payback or full-project price established.

Why not cache every answer and route everything to the cheapest model?

Change one lever, compare, then keep or revert

OPTIMISATIONS · ON THE ROADMAP



CACHE

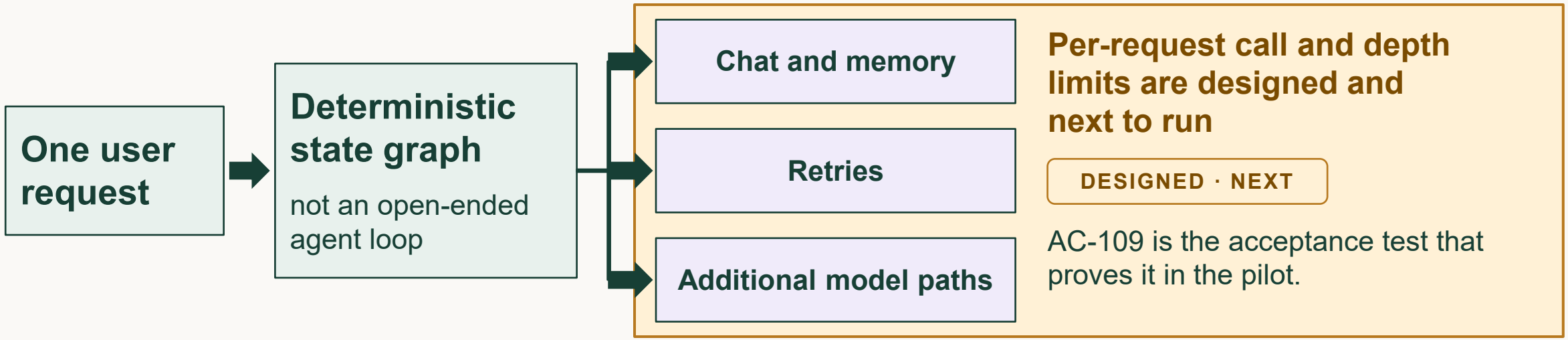
Same permission, evidence version and task validity required

ROUTING

Smallest model must clear held-out quality and latency thresholds

Sources: E19 · E04 · E09

What stops chat and agent sprawl multiplying hidden calls?



Existing bounded authority expires and exhausts

Finite calls, tokens and expiry per authority
Money caps come next

DESIGNED · NEXT

Complete per-call metering, request attribution, task and period money budgets, and a workflow-only stop

Amber box with outline = designed bound, next milestone · Purple = bounded model call · Green = existing workflow and authority

Sources: E19 · E09